TEST ANALYSIS REPORT · AUTOMATION QA

ParaBank Automation Framework

End-to-end UI test suite · Selenium + TestNG + Java 17 · Run: June 15, 2026, 11:45 PM

✓ 5/5 Tests Passed | 100% Pass Rate | ⏱ Total Duration: ~31s | ⊕ Chrome · ParaBank Demo

EXECUTION SUMMARY

TOTAL TESTS
5
All test cases run

PASSED
5
Zero failures

FAILED / SKIPPED
0
No errors detected

PASS RATE
100%
Full coverage pass

TOTAL DURATION
~31.8s
End-to-end wall clock

TEST RESULTS

Execution Results — All Tests

5 tests · Jun 15, 2026

●Passed **5**
●Failed **0**
●Skipped**0**

#	TEST NAME	MODULE / CLASS	PRIORITY	DURATION	STATUS
1	`verifyRegistration` New user registration with form data from Excel	RegistrationTest	P1	3.168s	•Pass
2	`verifyLogin` Username + password login flow validation	LoginTest	P2	4.074s	•Pass
3	`verifyAccountOverview` Post-login account dashboard assertion	AccountViewTest	P3	4.756s	•Pass

4	**verifyTransferFunds** Fund transfer login entry (data-driven)	TransferFundsTest	P5	4.168s	•Pass
5	**verifyBillPayment** Full bill pay flow with Excel data + assertion	BillTest	P6	14.643s	•Pass

EXECUTION TIMELINE

Sequential Run Order

11:46 PM – 11:47 PM

`verifyRegistration`
11:46:07 PM → 11:46:10 PM
3.168s

`verifyLogin`
11:46:17 PM → 11:46:21 PM
4.074s

`verifyAccountOverview`
11:46:27 PM → 11:46:32 PM
4.756s

`verifyTransferFunds`
11:46:44 PM → 11:46:48 PM
4.168s

`verifyBillPayment`
11:47:09 PM → 11:47:23 PM
14.643s

TECHNOLOGY STACK

Framework Architecture & Dependencies

pom.xml

CORE TESTING

Selenium WebDriver
4.34.0

TestNG
7.11.0

Cucumber (declared)
7.23.0

WebDriverManager
6.3.2

UTILITIES & REPORTING

ExtentReports
5.1.2

Apache POI
5.4.1

Log4j2
2.25.1

Java / Maven

17 / Surefire 3.5.3

FRAMEWORK STRUCTURE

Package Design — Page Object Model

LAYER	PACKAGE / CLASS	RESPONSIBILITY
•Base	`base.BaseFile`	Browser lifecycle: setup/teardown, driver init, multi-browser support (Chrome, Edge, Firefox)
•Pages	`pages.LoginPage / RegistrationPage / BillPage / AccountViewPage / TransferFundsPage`	POM encapsulation — element locators and page actions per screen
•Tests	`testcases.LoginTest / RegistrationTest / AccountViewTest / TransferFundsTest / BillTest`	TestNG @Test methods with data-driven execution via DataProviders
•Utilities	`utilities.DataProviders / ExcelUtils / ConfigReader / ExtentManager / ScreenshotUtils / WaitUtils`	Shared helpers: Excel I/O, config props, screenshot capture, explicit waits, report management
•Listeners	`listeners.TestListener`	ITestListener hooks for auto-screenshot on failure and Extent logging
•Config	`resources/config.properties · log4j2.xml · testdata/ParabankData.xlsx`	URL, browser, wait settings; log appenders; Excel test data (Login, Registration, BillPay sheets)

QA OBSERVATIONS

✓ STRENGTH — PASS RATE

All 5 test cases passed with zero failures across 3 consecutive execution runs logged. The framework demonstrates stable, repeatable results against the ParaBank demo environment.

✓ STRENGTH — POM DESIGN

Clear separation of concerns with Page Object Model, dedicated utility classes, and a centralized BaseFile for driver management. Easy to maintain and scale.

⚠ GAP — TRANSFERFUNDS ASSERTION

The `verifyTransferFunds` test only performs login steps — no actual transfer action or assertion verifies the transfer outcome. The test is incomplete for its stated purpose.

⚠ GAP — REGISTRATION ASSERTION

The `verifyRegistration` test lacks a success assertion after form submission. There is no check that a success message or redirect occurred post-registration.

i INFO — BILL PAY SPEED

Bill payment takes 14.6s vs ≈4s for all other tests. This is driven by an explicit `Thread.sleep(5000)` in `BillTest`. Replacing it with an explicit wait would improve reliability and speed.

i INFO — CUCUMBER DECLARED, UNUSED

Cucumber 7.23.0 is declared as a dependency in `pom.xml` but no feature files or step definitions exist. This adds unnecessary compile weight if BDD is not planned.

⚡ RISK — NO NEGATIVE TEST CASES

The suite covers only happy-path scenarios. No invalid-credential, insufficient-funds, or validation-error tests exist. Edge cases in login, registration, and bill pay are untested.

⚡ RISK — STATIC

BaseFile.driver is a static shared field. Running tests in parallel would cause race condition management.

RECOMMENDATIONS

1 Complete TransferFunds and Registration assertions

Add post-action assertions to confirm the transfer transaction or registration success pa without verifying the core business outcome.

2 Replace Thread.sleep() with explicit waits

Replace the hardcoded 5-second sleep WebDriverWait ExpectedConditions.textToBePresentInElement() f Payment Complete". This cuts runtime and avoids flakiness on slow

3 Add negative test

Implement invalid login (wrong password), duplicate registration, and bill pay with invalid ac increases the defect detection value of

4 Migrate driver to ThreadLocal for parallel

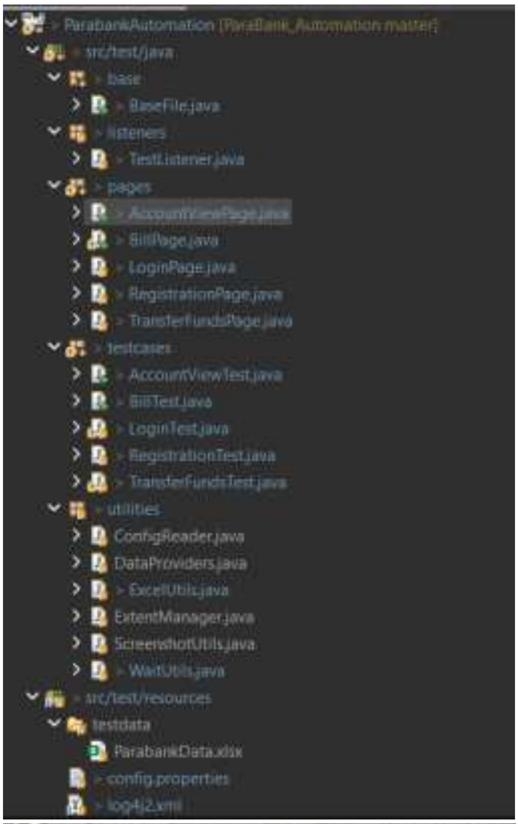
Replace static WebDriver driver in ThreadLocal<WebDriver> to safely support parallel TestNG executio total suite runtime significantly.

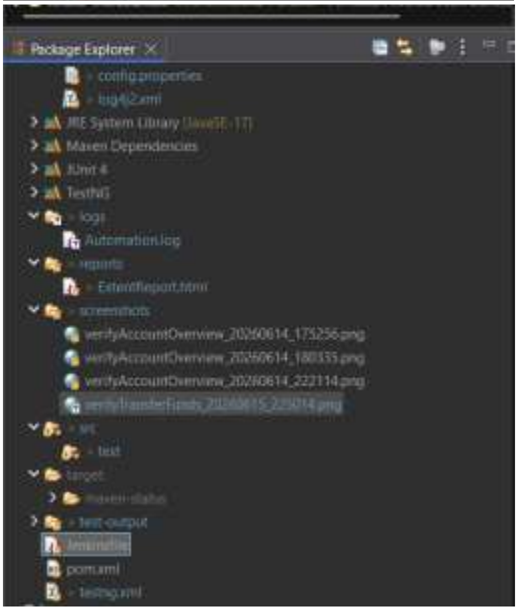
5 Remove or implement Cucumber

If BDD is not planned, remove the Cucumber dependency from pom.xml. If BDD ad definition classes to complete the Cucumber

6 Configure Jenkins pipeline with parameterized browser

The Jenkinsfile already exists. Enhance it browser to parameter at build time so Chrome, Firefox, triggered independently for cross-browser coverage.


ParabankAutomation [ParaBank_Automation master]
src/test/java
base
BaseFile.java
listeners
TestListener.java
pages
AccountViewPage.java
BillPage.java
LoginPage.java
RegistrationPage.java
TransferFundsPage.java
testcases
AccountViewTest.java
BillTest.java
LoginTest.java
RegistrationTest.java
TransferFundsTest.java
utilities
ConfigReader.java
DataProviders.java
ExcelUtils.java
ExtentManager.java
ScreenshotUtils.java
WaitUtils.java
src/test/resources
testdata
ParabankData.xlsx
config.properties
log4j2.xml


Package Explorer
config.properties
log4j2.xml
JRE System Library [JavaSE-17]
Maven Dependencies
JUnit 4
TestNG
logs
Automation.log
reports
ExtentReport.html
screenshots
verifyAccountOverview_20260614_175256.png
verifyAccountOverview_20260614_180335.png
verifyAccountOverview_20260614_222114.png
src
test
target
test-output
pom.xml
testng.xml

```
[RemoteTestNG] detected TestNG version 7.11.0
------ DATA FROM EXCEL ------
Payee Name : WaterBill
Address    : Mass Market
City       : Kolkata
State      : West Bengal
Zip        : 700102
Phone      : 9732385471
Account No : 16632
Amount     : 20
==============================
Experience the difference
Solutions
About Us
Services
Products
Locations
Admin Page
```

```
Experience the difference
Solutions
About Us
Services
Products
Locations
Admin Page
home
about
contact
Welcome Soonak Roy
Account Services
Open New Account
Accounts Overview
Transfer Funds
Bill Pay
Find Transactions
Update Contact Info
Request Loan
Log Out
Bill Payment Complete
Bill Payment to WaterBill in the amount of $20.00 from account 18888 was successful.
See Account Activity for more details.
Home|
About Us|
Services|
Products|
Locations|
Forum|
Site Map|
Contact Us
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===============================================
Suite
Total tests run: 5, Passes: 5, Failures: 0, Skips: 0
===============================================
```
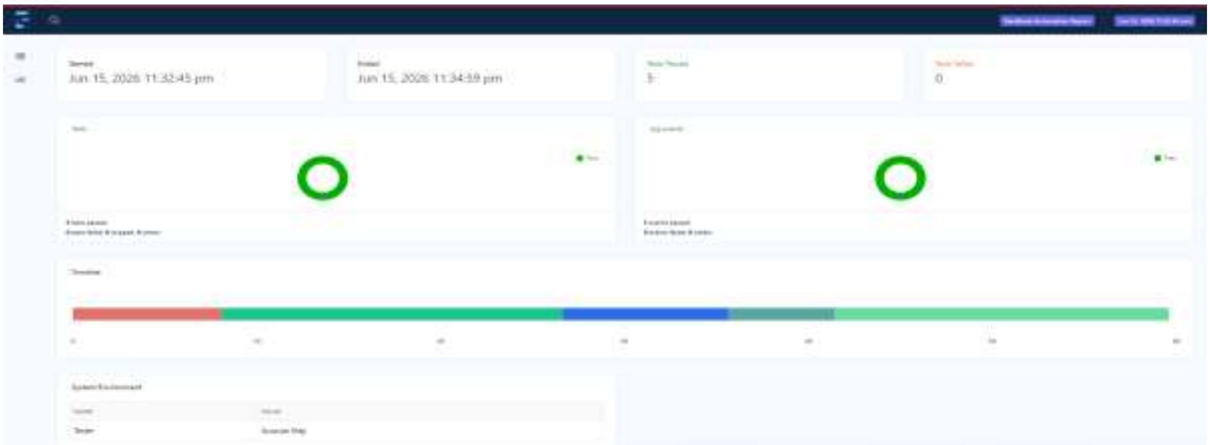
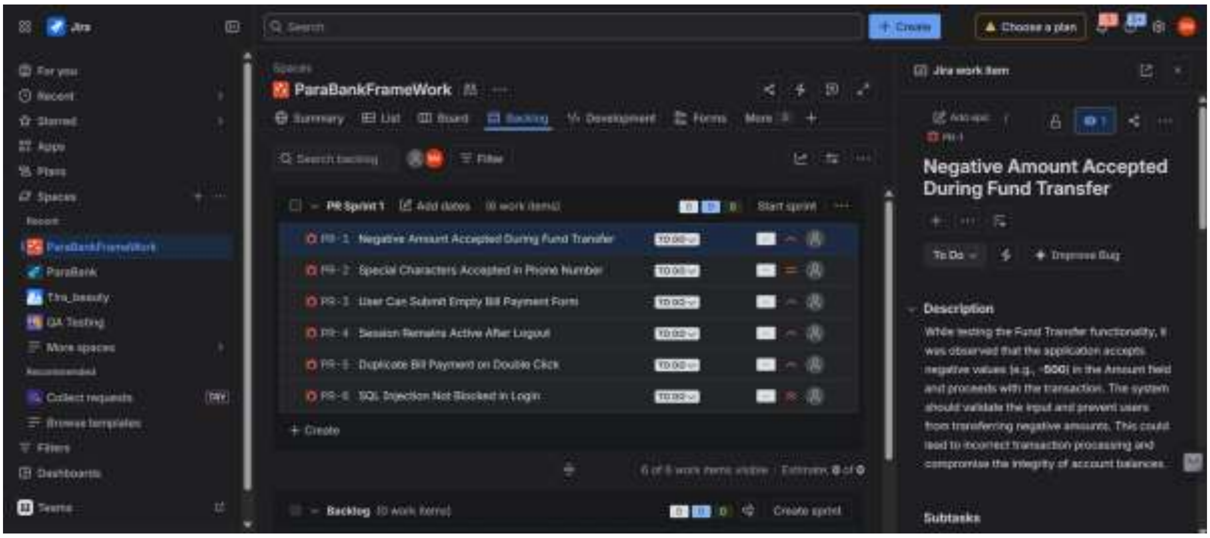
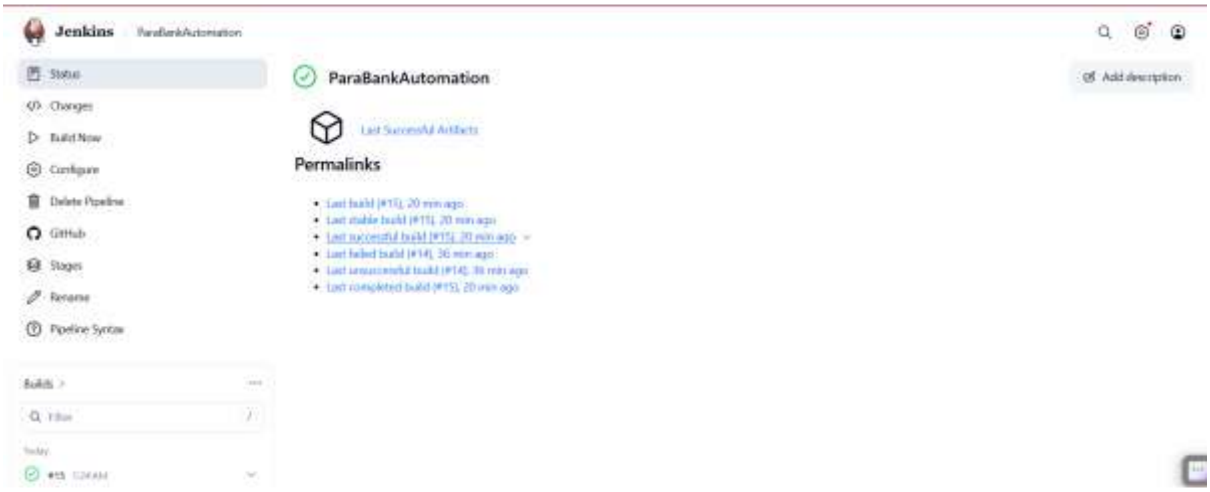





ParaBankFrameWork
Negative Amount Accepted During Fund Transfer
Special Characters Accepted in Phone Number
User Can Submit Empty Bill Payment Form
Session Remains Active After Logout
Duplicate Bill Payment on Double Click
SQL Injection Not Blocked in Login
Description
Subtasks



Jenkins
ParaBankAutomation
Permalinks